C:\hpc-23-feb-2026\.\21\eg159.c

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
void * do_something(void *ptr)
{
 int *w=(int *)ptr;
 long j=0;
 while(*w)
 {
  j++;
  printf("%ld\n",j);
  sleep(1);
 }
 return NULL;
}
int main()
{
 pthread_t tid;
 int keep_working=1;
 pthread_create(&tid,NULL,do_something,(void *)&keep_working);
 sleep(5);
 keep_working=0;
 pthread_join(tid,NULL);
 return 0;
}
```

C:\hpc-23-feb-2026\.\21\eg160.c

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
void * do_something(void *ptr)
{
 long j=0;
 while(1)
 {
  j++;
  printf("%ld\n",j);
  sleep(1);
 }
 return NULL;
}
int main()
{
 pthread_t tid;
 pthread_create(&tid,NULL,do_something,NULL);
 sleep(5);
 pthread_cancel(tid);
 pthread_join(tid,NULL);
 return 0;
}
```

C:\hpc-23-feb-2026\.\21\eg161.c

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
void * do_something(void *ptr)
{
 long j=0;
 while(1)
 {
  j++;
  printf("%ld\n",j);
 }
 return NULL;
}
int main()
{
 pthread_t tid;
 pthread_create(&tid,NULL,do_something,NULL);
 sleep(5);
 pthread_cancel(tid);
 pthread_join(tid,NULL);
 return 0;
}
```

C:\hpc-23-feb-2026\.\21\eg162.c

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
void * do_something(void *ptr)
{
 long j=0;
 while(1)
 {
  j++;
  printf("%ld\n",j);
  pthread_testcancel();
 }
 return NULL;
}
int main()
{
 pthread_t tid;
 pthread_create(&tid,NULL,do_something,NULL);
 sleep(5);
 pthread_cancel(tid);
 pthread_join(tid,NULL);
 return 0;
}
```

C:\hpc-23-feb-2026\.\21\eg163.c

```
#include<stdio.h>
#include<stdlib.h>
typedef struct _stack_node
{
 int num;
 struct _stack_node *next;
}stack_node;
typedef struct _stack
{
 stack_node *top;
}stack;
void stack_push(stack *s,int num)
{
 stack_node *node;
 node=(stack_node *)malloc(sizeof(stack_node));
 node->num=num;
 node->next=s->top;
 s->top=node;
}
// I am not writing top
void stack_pop(stack *s)
{
 stack_node *node;
 if(!(s->top)) return;
 node=s->top;
 s->top=node->next;
 free(node);
}
void some_func()
{
 stack stk;
 stk.top=NULL;
 stack_push(&stk,10);
 stack_push(&stk,20);
}
int main()
{
 some_func();
 return 0;
}
```

C:\hpc-23-feb-2026\.\21\eg164.c

```
#include<stdio.h>
#include<stdlib.h>
typedef struct _stack_node
{
 int num;
 struct _stack_node *next;
}stack_node;
typedef struct _stack
{
 stack_node *top;
}stack;
void stack_push23423(stack *s,int num)
{
 stack_node *node;
 node=(stack_node *)malloc(sizeof(stack_node));
 node->num=num;
 node->next=s->top;
 s->top=node;
}
// I am not writing top
void stack_pop45645(stack *s)
{
 stack_node *node;
 if(!(s->top)) return;
 node=s->top;
 s->top=node->next;
 free(node);
}
#define stack_push(a,b)\
{\
 stack_push23423(a,b)
#define stack_pop(a)\
 stack_pop45645(a);\
}
void some_func()
{
 stack stk;
 stk.top=NULL;
 stack_push(&stk,10);
 printf("Cool\n");
 stack_pop(&stk);
}
int main()
{
 some_func();
 return 0;
}
```

C:\hpc-23-feb-2026\.\21\eg165.c

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
void some_function_to_release_resources(void *ptr)
{
 printf("SOME FUNCTION TO RELEASE RESOURCES GOT CALLED\n");
}
void * do_something(void *ptr)
{
 // following is a macro
 pthread_cleanup_push(some_function_to_release_resources,NULL);
 long j=0;
 while(1)
 {
  j++;
  printf("%ld\n",j);
  pthread_testcancel();
 }
 printf("We are out of the loop\n");
 pthread_cleanup_pop(0);
 return NULL;
}
int main()
{
 pthread_t tid;
 pthread_create(&tid,NULL,do_something,NULL);
 sleep(5);
 pthread_cancel(tid);
 pthread_join(tid,NULL);
 return 0;
}
```

C:\hpc-23-feb-2026\.\21\eg166.c

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
void some_function_to_release_resources(void *ptr)
{
 printf("SOME FUNCTION TO RELEASE RESOURCES GOT CALLED\n");
}
void * do_something(void *ptr)
{
 // following is a macro
 pthread_cleanup_push(some_function_to_release_resources,NULL);
 long j=0;
 while(1)
 {
  j++;
  printf("%ld\n",j);
  pthread_testcancel();
  break;
 }
 printf("We are out of the loop\n");
 pthread_cleanup_pop(0);
 return NULL;
}
int main()
{
 pthread_t tid;
 pthread_create(&tid,NULL,do_something,NULL);
 sleep(5);
 pthread_cancel(tid);
 pthread_join(tid,NULL);
 return 0;
}
```

C:\hpc-23-feb-2026\.\21\eg167.c

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
void some_function_to_release_resources(void *ptr)
{
 printf("SOME FUNCTION TO RELEASE RESOURCES GOT CALLED\n");
}
void * do_something(void *ptr)
{
 // following is a macro
 pthread_cleanup_push(some_function_to_release_resources,NULL);
 long j=0;
 while(1)
 {
  j++;
  printf("%ld\n",j);
  pthread_testcancel();
  break;
 }
 printf("We are out of the loop\n");
 pthread_cleanup_pop(1);
 return NULL;
}
int main()
{
 pthread_t tid;
 pthread_create(&tid,NULL,do_something,NULL);
 sleep(5);
 pthread_cancel(tid);
 pthread_join(tid,NULL);
 return 0;
}
```

C:\hpc-23-feb-2026\.\21\eg168.c

```
#include<stdio.h>
#include<unistd.h>
int main()
{
 printf("%ld cores\n",sysconf(_SC_NPROCESSORS_ONLN));
 return 0;
}
```